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Evaporation and the development of the local boundary layer over an irrigated surface in an arid region

J.H. Prueger [a,*], L.E. Hipps [b], D.I. Cooper [c]

[a] *National Soil Tilth Laboratory, 2150 Pammel Drive, Ames, IA 50011-4420, USA*
[b] *Utah State University, Logan, UT 84322, USA*
[c] *Los Alamos National Laboratories, Los Alamos, NM 87544, USA*

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Abstract

Heterogeneous landscapes common to arid and semi-arid regions are often characterized by a horizontal advection of sensible heat from dry surfaces to wet surfaces which modifies the evaporation and water balance of moist areas.

This study centered around a well-watered alfalfa field surrounded locally by arid surfaces, but within a large complex of irrigated fields where both local and regional scale advection were present. Latent and sensible heat flux densities were measured using eddy correlation at four different locations within the same field. Tethered balloons were used to obtain vertical profiles of temperature, humidity, wind speed and direction from the surface to heights of 50 m at the leading and downwind edges of the field.

Results show that sensible heat flux and turbulence intensity were strongly correlated indicating the dependence of sensible heat transfer on turbulence. Evaporation in excess of the equilibrium rate was governed by turbulence intensity and the advection of sensible heat. Growth of the local boundary layer above the alfalfa canopy as defined by temperature and humidity profiles averaged about 30 m over a 450 m traverse. The height of the developed local boundary layer was strongly correlated to the intensity of turbulence. Short-term evaporation rates were calculated from the upwind and downwind profiles of humidity and wind, using a modified vapor budget technique, and found to average approximately 10% lower than the measured evaporation rates. The contribution to evaporation from the advection of sensible heat was also calculated from the upwind and downwind profiles and found to range between 28–90% of the total evaporation.

* Corresponding author.

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1. Introduction

Understanding evaporation of water from the earth's surface in a semi-arid region can provide a valuable contribution to understanding the local hydrology of water-limited areas. In general, net radiation is the primary source of energy for evaporation of water from soil and leaf surfaces. This is most certainly the case in areas where climate can be classified as humid and the surface of the earth can be treated as largely homogeneous with respect to water availability from both soil and vegetative surfaces.

In arid and semi-arid regions however, the surface is largely heterogeneous in which irrigated fields or wet patches are often found lying adjacent to arid landscapes. The advection of sensible heat from dry surfaces can be a significant source of energy that must be taken into consideration when evaluating evaporation in arid and semi-arid regions. The term advection as used in this study will be synonymous with the transport of energy or mass in the horizontal plane in a downwind direction by wind. The terms local and regional advection are used to identify sources of preconditioned air. Regional advection occurs when horizontal transport of sensible heat energy is the result of movement of warm dry air masses on a synoptic scale (Blad and Rosenberg, 1974). An appropriate example of local advection would be an irrigated field crop bordered by a large area of dry fallow land or an airport tarmac.

Local advection is often associated with heterogeneous landscapes commonly found in irrigated agricultural regions. More often than not local and regional advection occur simultaneously. Ordinary energy balance methods generally assume the contribution from advection to be negligible. Lang (1973) proposed a simple first order correction used with a Bowen ratio (Bowen, 1926) function to account for the contribution from advection in semi-arid climates. The model accounts for the divergence of sensible heat flux between a surface and a height z under advective conditions and is a two-dimensional energy balance equation. Data from Rider et al. (1963) were used to test the model in which the results suggested that a two-dimensional energy balance approach may allow the calculation of latent heat flux under intense conditions of local advection.

In a two part study (McNaughton, 1976a; McNaughton, 1976b) a model was developed which successfully reproduced the equilibrium evaporation result that had been previously observed in several studies (Denmead and McIlroy, 1970; Davies, 1972; Wilson and Rouse, 1972). In Part I McNaughton introduced a general expression for evaporation at all x consisting of an 'equilibrium evaporation' term and a term to account for the contribution of evaporation from advection.

In Part II the contribution to evaporation from advection was investigated for a simple case of steady-state evaporation downwind of a physical boundary separating two surfaces having different resistances and available energies.

Advected sensible heat energy originating from a dry source upwind from a well watered surface increases the amount of energy available for evaporation as well as increases the vapor pressure gradient between the advected parcel of air and the cooler moist vegetative surface. As this warmer and drier parcel of air is transported downwind over an evaporating surface, part of its energy will exchange with the moist surface that will result in increased evaporation which produces a horizontal gradient of temperature

and water vapor content of the air immediately above a well watered surface between the leading and downwind edges of a field (Oke, 1978).

The effect of advection can be considerable. Hanks et al. (1971) reported errors up to 45% in estimated evapotranspiration for measurements made over irrigated crops in a semi-arid region. A major part of this error was attributed to the vertical flux divergence which occurs under advective conditions. In arid and semi-arid regions, evaporation by a well watered vegetative surface has been frequently found to exceed the energy content of net radiation by sometimes as much as a factor of two or more (Brakke et al., 1978). Advected heat energy from warm dry surroundings can have a profound impact on the local hydrology of an irrigated region. Previous studies have focused on a shallow layer of air within a few meters of the surface. However, the process by which advection distorts the surface energy balance involves a much deeper layer of air. Measurements of surface fluxes and properties of the boundary layer should be made simultaneously to examine this phenomenon in a more integrated fashion. This study examines the evaporation and local boundary layer over an irrigated alfalfa field bordered by arid landscapes. The specific objectives are:

1. determine short term evaporation rates and their spatial and temporal variation from irrigated alfalfa under advective conditions;
2. examine the relationship of boundary layer growth and properties and strength of advection of heat, with turbulence intensity as well as calculate the evaporation rate and the contribution of sensible heat advection from horizontal changes in the vertical profiles of temperature, humidity, and wind inside the local boundary layer.

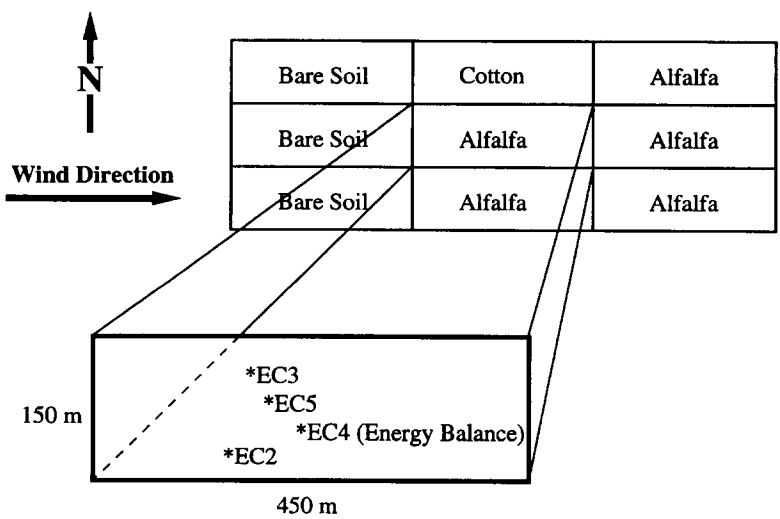

Fig. 1. Experimental layout located at the Maricopa Agricultural Center, Maricopa, AZ. The mean wind direction for the study was from the west.

2. Materials and methods

2.1. Study area

This study was conducted at the Maricopa Agricultural Research Center located near Phoenix, AZ, during the last week of June, 1990. There are a mosaic of irrigated fields at this location which are surrounded by an expanse of desert landscape. An alfalfa field of uniform canopy was selected, the dimensions of the field were 150 m wide by 450 m long with the length of the field oriented in an east–west direction, Fig. 1. From the west, the experiment field was bordered by a dry fallow field, to the east were more irrigated alfalfa fields, along the north edge were fields planted to cotton, and to the south were more alfalfa fields. All fields bordering the study site were of similar area (150 × 450 m). All of the alfalfa fields had been surface irrigated two days prior to the beginning of the study and were characterized as having maximum water availability for evaporation.

2.2. Surface flux measurements

Surface flux measurements were made using 4 eddy correlation stations that were staggered in position as shown in Fig. 1. In addition, one of these stations was equipped as a full energy balance station, and was positioned near the center of the field. Each eddy correlation station consisted of a 1-dimensional sonic anemometer (Campbell Scientific Instruments [1], Logan, UT), fine wire thermocouple and a fast response krypton hygrometer. The instruments were positioned at a height of 1.6 m above the alfalfa canopy and located 160 m from the leading edge. The height of the canopy was 50 cm above the soil surface. The energy balance station included the above mentioned instruments as well as a Schenk[1] net radiometer, and 2 soil heat flux plates with averaging soil temperature probes located 0–8 cm below the soil surface. Windspeed and direction were measured using a cup anemometer and wind vane.

A Campbell Scientific 21x[1] was used to sample the eddy correlation instruments at a frequency of 10 Hz, after which latent and sensible heat flux values were calculated for 10 min block intervals and averaged as 30 min fluxes. Net radiation and soil heat flux were measured at 2 s intervals and averaged for a 30 min output. Soil heat flux values were corrected for storage in the upper soil layer using soil temperature, moisture, and bulk density data. The windspeed and direction were sampled at 60 s intervals and averaged for 30 min output. Continuous data collection began on June 27 at 1400 h and continued through June 29 until 1730 h.

2.3. Eddy correlation instrument variability

Prior this study instrument variability was assessed by comparing ***LE*** and ***H*** fluxes measured with a set of eddy correlation systems including those used in the study which

[1] The USDA does *not* endorse any product or company. Trade and company names are provided only for facilitation purposes for the reader.

were collocated over a surface of irrigated pasture grass in Logan, UT. Results from this comparison showed average variations for ***LE*** and ***H*** measurements of about 7% (Swiatek et al. unpublished data).

2.4. Temperature and vapor pressure profiles

Vertical air temperature and water vapor profiles were measured using two tethered balloon systems operated simultaneously at opposite ends of the field.

The first system was located just upwind of the leading edge of the field, which represented a boundary between a well-watered alfalfa surface and a dry fallow field, a second system above the alfalfa, about 450 m downwind from the leading edge.

The balloon-borne instrument package measured wet and dry bulb temperatures, atmospheric pressure, windspeed, and direction. The flights were performed in the afternoon between 1500 and 1630 h on June 28 and 29 when windspeed, turbulence, and advection were most pronounced. A typical measuring event of the tethered systems consisted of a simultaneous lift-off of the balloons beginning at approximately 1.5 m above the surface to a height of approximately 50 m. At 50 m the balloons were maintained stationary for approximately 3 min after which the decent portion of the launch was initiated. Only ascent data were used in the analyses. A total of seven launches were made over the two day period. Data from these measurements were used to calculate vertical profiles of potential temperature and specific humidity which were further processed using a cubic spline smoothing technique (Kimball, 1976) to account for fluctuations caused by turbulence.

Estimates of average evaporation from the field as well as the contribution of advection to evaporation were estimated from upwind and downwind profiles of temperature, humidity, and windspeed. The method used to estimate evaporation from a control volume over the alfalfa field is similar to that described by Sill et al. (1984) and Hipps and Zehr (1995). The approach essentially utilizes a water–vapor budget method to account for water vapor as it flows into and out of a control volume of the system being investigated. A similar approach was used for estimating the amount of evaporation due to the contribution from local sensible heat generated over a dry fallow surface.

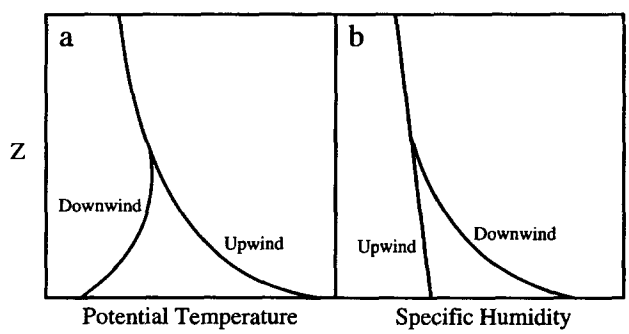

Fig. 2. Theoretical adjustments of vertical profiles of (a) potential temperature (θ), and (b) specific humidity (q) downwind of a leading edge.

2.5. Vapor budget method

The water–vapor budget method is based on a control volume, which is defined by upwind and downwind boundaries of the field as well as the height at which water vapor profiles from the boundaries converge (Fig. 2).

The amount of water vapor that is added to a parcel of warm dry air entering a control volume of air over an evaporating surface is governed by the distance that the parcel of air must travel through the control volume as well as the time required for the parcel to exit the control volume (residence time). The residence time can be easily calculated from the mean wind velocity and the distance between upwind and downwind boundaries of the field. Following Sill et al. (1984) and Hipps and Zehr (1995) the mean evaporation rate of the surface inside the control volume is calculated as:

$$E_{avg} = \frac{L}{X}\int_0^Z U\rho_a(z)[q_d(z) - q_u(z)]\,dz \tag{1}$$

where E_{avg} is the evaporation rate in W m^{-2}, over a finite downwind fetch X (m) of the control volume, L the latent heat of vaporization (J KG^{-1}), Z the height (m) at which the specific humidity profiles measured from both upwind and downwind boundaries converge, U the mean wind velocity (m s^{-1}) through the control volume, ρ_a is density of moist air (kg m^{-3}) at any given height z, and q_d and q_u (dimensionless) are specific humidities at any given height z downwind and upwind, respectively. In this form, Eq. 1 describes the amount of water vapor input to a control volume from an underlying surface which is equal to the difference between the water vapor mass entering the control volume at the upwind position, and the amount of water vapor mass exiting the control volume at the downwind position.

In a similar manner, the contribution to latent heat flux from horizontal advection of sensible heat through a control volume can be calculated from the depletion of sensible heat content in the control volume and is expressed as:

$$E_{adv} = \frac{C_p}{x}\int_0^Z U\rho_a(z)[\theta_u(z) - \theta_d(z)]\,dz \tag{2}$$

where E_{adv} is in W m^{-2} and represents evaporation resulting from the contribution of advection, C_p the specific heat of moist air (J kg^{-1} K^{-1}), θ_u and θ_d are potential temperatures (K) at any given height z upwind and downwind, respectively, Z the height of a developed temperature inversion (m), and ρ_a, U, and x have been previously defined in Eq. 1. In this form, Eq. 2 calculates the amount of thermal energy that is removed from a parcel of hot dry air as it traverses a transpiring surface. The amount of internal energy removed from the control volume by evaporation is related to the area between the upwind and downwind vertical temperature profiles. Two important assumptions are made when using this method; first, the net flux through the top of the control volume is zero, and second, that lateral (Y direction) net transport is also zero.

3. Results and discussion

3.1. Energy balance closure

Surface energy balance closure provides an indication of self-consistency and uncertainty of the energy balance measurements. It is computed as $(H + LE)/(R_n - G)$. In

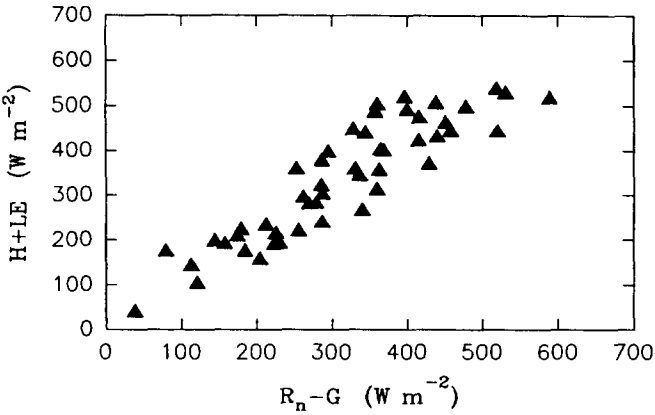

Fig. 3. 30-min energy balance closures for periods of positive net radiation for June 27–29, 1990. Overall closure which was calculated from the sum of all energy components was 0.93.

theory, if all measurements were without error the closure would be unity and deviations from unity would suggest the amount of measurement uncertainty. Fig. 3 shows 30-minute results of $H+LE$ plotted against $R_n - G$ for periods of positive net radiation (daytime) for the two day study. Calculation of overall energy closure for this period was accomplished by summing each individual 30-minute value of R_n, G, LE, and H, and then calculating closure as a total of each energy component for the two day period. This resulted in an overall closure of 0.93, which represents excellent internal self-consistency of measurements of the energy balance components.

3.2. Eddy correlation results

Latent heat fluxes from 0600 to 1200 h on June 29 are shown in Fig. 4(a). The environmental conditions during this period were characterized by low windspeeds (0–1 m s^{-1}) and clear skies. The results of sensible heat flux (H) and the standard deviation of the vertical wind velocity component (σ_w), an indication of turbulence intensity, are shown for the same day and time periods in Fig. 4(b) and 4(c), respectively.

In general, fluctuations in LE, H, and σ_w were only moderately correlated during the morning hours. Apparent spatial variations in LE began to appear after 0900 h and increased in magnitude as the morning hours progressed. Sensible heat flux densities showed less variability then LE during the morning hours. Sensible heat fluxes varied by as much as 35 W m^{-2} from one station to another (Fig. 4(b), 0930 h) while latent heat fluxes appeared to vary by as much as 150 W m^{-2} (0930 h). Spatial variability of σ_w during the morning hours was small (Fig. 4(c)). The very low windspeeds (0–1 m s^{-1}) and variable wind directions during the morning hours suggest conditions may have been non-steady state. Thus the uncertainty of the flux measurements under such conditions make it difficult to determine if the apparent variations were real.

During the afternoon hours from 1400 through 1730 h LE flux values increased significantly, Fig. 5(a). These increases were associated with increased wind velocities

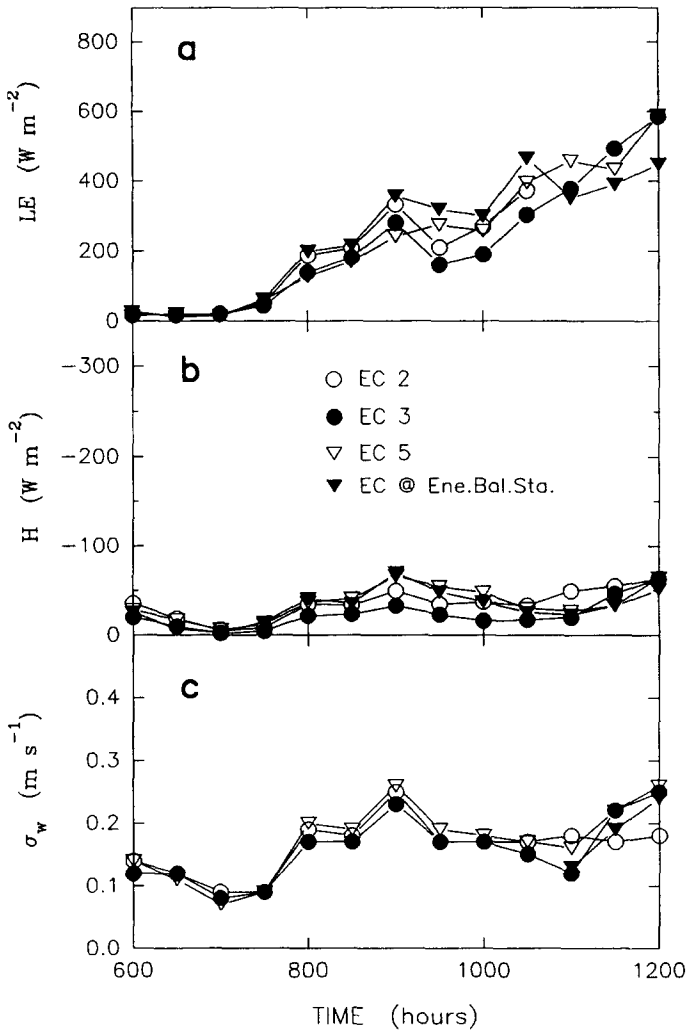

Fig. 4. (a–c) 30-min average *LE*, *H*, and σ_w data, respectively, for June 28 for the hours of 0600 through 1200. Environmental conditions were characterized by low winds (1–2 m s^{-1}) and air temperatures above 35°C.

(7–10 m s^{-1}) and upwind air temperatures near 50°C. Sensible heat and turbulence intensity values presented in Fig. 5(b) and 5(c) are also seen to increase significantly. Temporal changes in *LE* values were did not correlate to those of *H* and σ_w. However, a strong correlation was observed between *H* and σ_w indicating a strong dependence of sensible heat transfer on turbulence intensity. Spatial variations in *LE* were insignificant. The larger horizontal wind velocities promoted strong advection of sensible heat from the fallow field. Fig. 5(b) shows an increase in spatial variability ($\cong$ 45–60 W m^{-2}, 1430–1700 h) for sensible heat fluxes compared with those observed during the

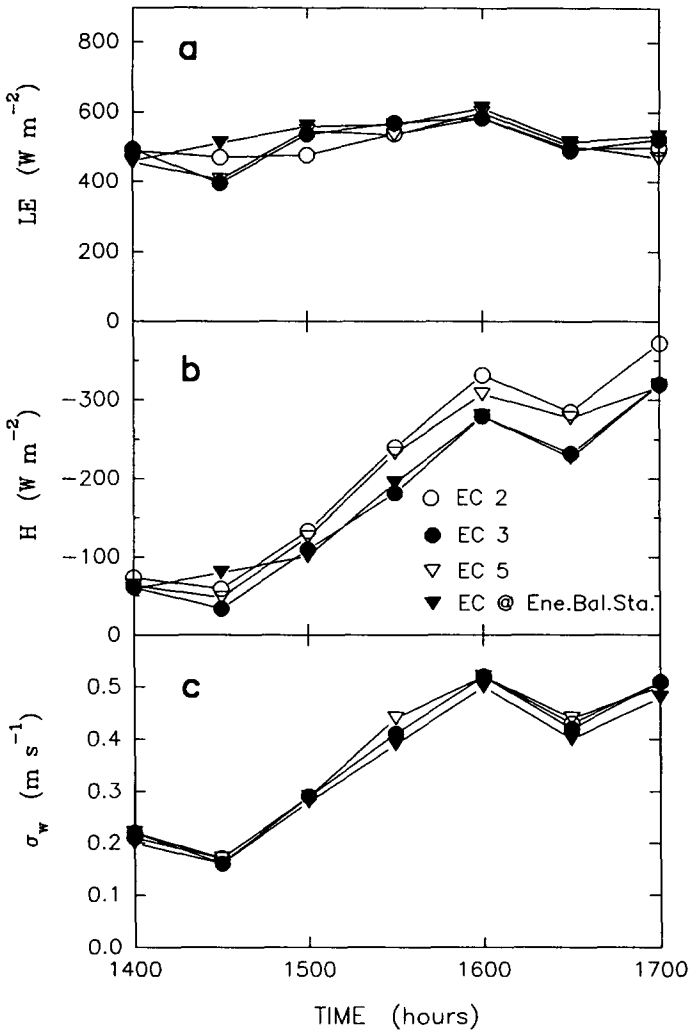

Fig. 5. (a–c) 30-min average *LE*, *H*, and σ_w data, respectively, for June 28 for the hours of 1400 through 1700. Conditions during this period were characterized by strong winds (7–10 m s^{-1}) and air temperatures above 50°C.

morning hours ($\cong$ 35 W m^{-2}, Fig. 4(b)). Significant increases in the magnitudes of *H* fluxes during 1400 to 1700 h are related to intense local advection. The values of σ_w between 1400 to 1730 h show little spatial variation, and were at least twice as large as those observed during the morning hours, Fig. 5(c). The more intense turbulence enhanced scaler exchanges between the alfalfa canopy and atmosphere. Although *LE* fluxes are not seen to be strongly correlated with *H* and σ_w values, it can be observed that σ_w and *H* are generally increasing throughout the afternoon while *LE* fluxes

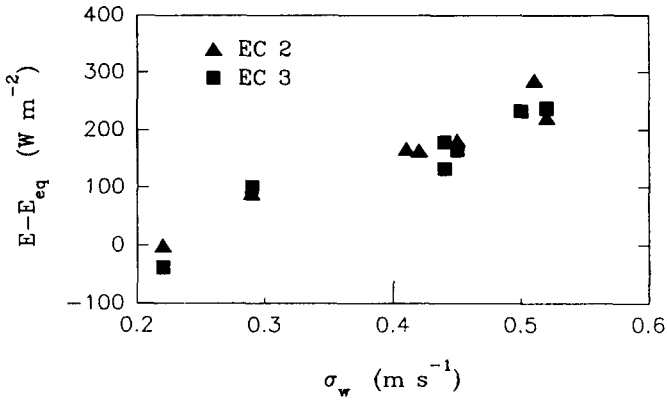

Fig. 6. The contribution to the LE flux from the advection of H, calculated as the difference between measured evaporation from eddy correlation stations 2 and 3 and the evaporation equilibrium (E_{eq}) calculated as measured $R_n - G$ as a function of turbulence (σ_w).

remain relatively constant despite rapidly decreasing $R_n - G$ values, thus demonstrating the effect of advection increasing as the afternoon progresses.

Under conditions in which an extensive vegetative surface is effectively decoupled from the atmosphere, the equilibrium evaporation rate from a surface was expressed by McNaughton and Jarvis (1983) as:

$$\lambda E = \frac{s(R_n - G)}{s + \gamma} \tag{3}$$

where λE is the latent heat of evaporation, G soil heat flux, s slope of the saturation vapor pressure function, and γ is the psychrometric constant. A more accurate derivation by Paw U and Gao (1988) yields:

$$E_{eq} = R_n - G \tag{4}$$

Under advective conditions a vegetative canopy is coupled to the atmosphere and receives additional energy by the advection of sensible heat, or expressed in another way, by the transport of saturation deficit from a hot dry surface to a cool well-watered surface. The contribution to latent heat flux from the advection of sensible heat can be expressed as the difference between the measured latent heat flux (E) and the equilibrium evaporation (E_{eq}) for the time period under consideration. Fig. 6 shows the relationship between $E - E_{eq}$ and turbulence intensity σ_w from two eddy correlation stations. The strong positive correlation between $E - E_{eq}$ and σ_w indicates that the contribution to evaporation by advection of sensible heat is strongly governed by the intensity of turbulence.

Fig. 7(a) shows latent heat values between 0600 and 1330 h for the previous day, June 28. The conditions of the environment for this day were nearly identical to those of June 29. The pattern of spatial variability of latent heat flux densities was similar to

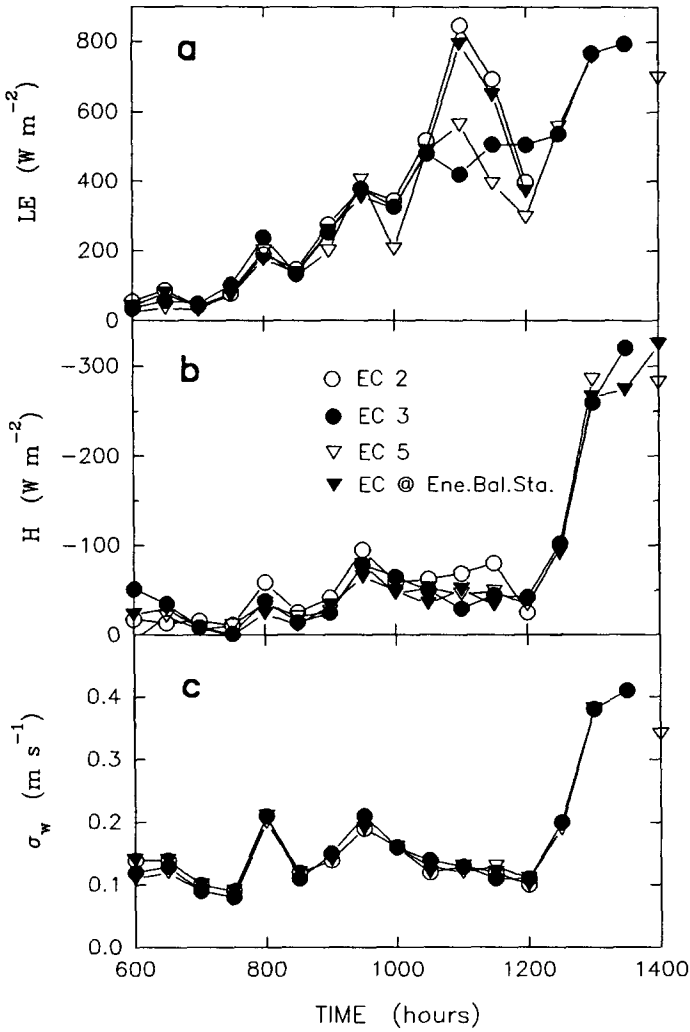

Fig. 7. (a–c). 30-minute average LE, H, and σ_w data respectively for June 27 during the hours of 0600 through 1400. Environmental conditions were nearly identical to those of June 28.

those observed on June 29, where apparent spatial variations were observed in the early morning hours which decreased in magnitude by late morning and early afternoon. Late afternoon measurements on June 28 were interrupted due to technical problems.

More spatial variability in H was observed during morning hours which had lower windspeeds (1–2 m s^{-1}). In general, it can be seen from Figs. 4–7 that temporal variations in LE are weakly associated with turbulence intensity σ_w, but $E - E_{eq}$ is governed by σ_w.

4. Tethersonde results

Fig. 8(a),(b) shows typical results for profiles of θ and q. A temperature inversion developed over the alfalfa field, which reached a height of approximately 30 m at the downwind site. The solid lines represent a cubic spline curve fit to actual measured data, and were calculated following Kimball (1976). The unstable profile at the upwind side reflects the positive sensible heat flux from the dry bare soil surface. The temperature inversion grows as evaporation removes internal energy from the air mass by means of turbulent exchange. The point of convergence of the two profiles indicates the depth of the layer of air that has been modified by the evaporating surface.

Specific humidity profiles presented in Fig. 8(b) show the accumulation of water vapor into the air mass as it moved downwind. As with the θ profiles the point of convergence of the q profiles was about 30 m. This shows symmetry in the turbulence exchanges of energy and vapor above the surface. Results from other launches were similar, although the depth of the boundary layer development varied somewhat. Fig. 9 presents height Z of the local boundary layer estimated from each of the tethersonde launches plotted against averaged σ_w's from the surface stations during the same period of time. This result shows a strong correlation between depth of the local boundary layer and intensity of turbulence generated over the alfalfa canopy. This result indicates that the depth of the local boundary layer over the evaporating alfalfa canopy was strongly dependent on the intensity of turbulence.

The evaporation rates during several launches were estimated from specific humidity data using Eq. 1. Table 1 shows a comparison of predicted ***LE*** fluxes with individual measured fluxes from the surface stations during the same time period that the tethersonde measurements were made. Such estimated ***LE*** fluxes were lower than all

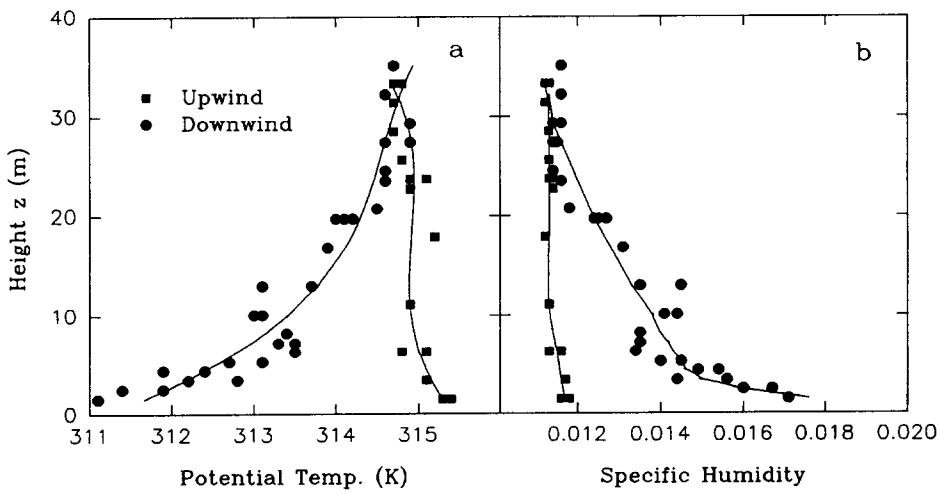

Fig. 8. (a,b) Typical potential temperature and specific humidity profiles from upwind and downwind positions of the alfalfa field for June 29, 1990 between 1511 and 1518 h. Conditions were characterized by temperatures of 50°C and strong winds from the west (7–10 m s^{-1}).

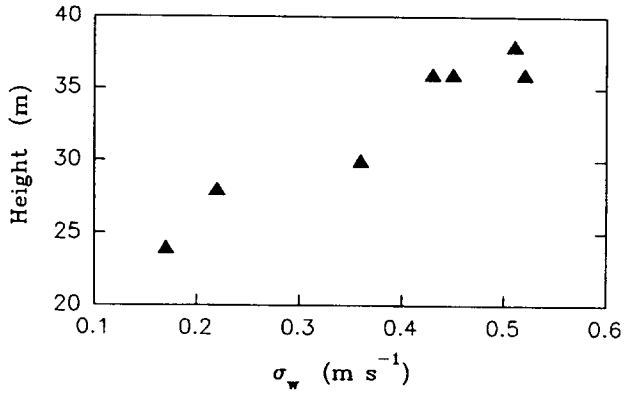

Fig. 9. Relationship between the height of the boundary layer over the alfalfa field and turbulence (σ_w) for June 29, 1990.

measured surface fluxes by an average of 10 percent which is reasonable. Table 2 shows the quantity of sensible heat content removed from the local boundary layer calculated from Eq. 2. This is an estimate of the evaporation due to advection. The results demonstrate that values ranging from 28–90% of the energy used for evaporation from the alfalfa canopy were provided by advection. These results compare favorably with those obtained by other methods of measuring evaporation under advective conditions (Hanks et al., 1971; Brakke et al., 1978; Hipps and Zehr, 1995).

Table 1
Estimated *LE* from two tethered balloon launches and measured *LE* from the eddy correlation stations for the same time period on 29 June

Date	Launch #	Measured *LE* (W m^{-2})				Avg. meas. *LE* (W m^{-2})	Est. *LE*$^{(W\ m-2}$)
		EC 2	EC 3	EB	EC 5		
June 29	1	536.0	546.7	558.5	546.8	547.0	458.8
June 29	3	582.7	598.9	610.9	519.1	577.9	511.0

Table 2
Estimated contribution to *LE* from the advection of sensible heat derived from tethered balloon data and average measured *LE* from the eddy correlation stations

Date	Launch #	Est. LE_{adv} (W m^{-2})	Avg. meas. *LE* (W m^{-2})	% of *LE*
June 28	1	417.8	571.6	71
June 28	2	311.9	605.3	52
June 29	1	151.3	547.0	28
June 29	3	419.7	577.9	73
June 29	4	441.3	491.6	90
June 29	5	457.4	485.1	86

5. Conclusions

The results demonstrate that apparent spatial variations in evaporation from a homogeneous field were more pronounced during the morning hours when conditions of weak sensible heat advection were prevalent. This was associated with low windspeeds and variable wind directions which reflected non-steady state conditions. Under such conditions uncertainty of the individual flux measurements make it difficult to interpret the apparent variations between stations during this time. Sensible heat flux and σ_w were strongly correlated indicating the dependence of sensible heat transfer to turbulence. Evaporation in excess of the equilibrium rate was closely governed by turbulence intensity and the advection of sensible heat flux.

Growth of the local boundary layer as defined by temperature and humidity profiles averaged about 30 m in height for the 450 m traverse. Development of the local boundary layer above the alfalfa canopy was dependent on the intensity of turbulence. The variation in depth was almost solely explained by σ_w. Despite a strong thermal inversion that developed over the evaporating canopy, the intensity of turbulence during advection was rather large. This is likely a result of the large scales of turbulent eddies generated over a dry landscape upwind of the field site.

Short-term rates of evaporation were estimated with a modified vapor budget technique using measurements of upwind and downwind profiles of humidity and wind speed. The estimated ***LE*** fluxes averaged overall approximately 10% lower than the measured fluxes. The contribution to evaporation solely due to advection of sensible heat was also estimated from upwind and downwind profiles of temperature and windspeed and found to range between 28–90% of the total evaporation. The contribution to evaporation from advection was very large, but exhibited a great deal of temporal variability related to that of turbulence.

Acknowledgements

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